Arduino-Based AC Voltage and Overcurrent Protection System

PROJECT OVERVIEW

This project simulates the load protection circuit using Arduino Uno for overvoltage, undervoltage and overcurrent condition inside Proteus 8 Software. The voltage protection is determined by the voltage divider circuit, and the overcurrent is measured by ACS712 sensor. Moreover, it also features a virtual terminal inside proteus so that user can change overvoltage, overcurrent and undervoltage threshold. However, user cannot put any value they wish to set as threshold value, there is a limit set for each value. Manual reset switch is used for the restart of the circuit.

Components Used

- Simulino (Arduino Uno)
- Alternator (AC supply)
- Transformer
- ACS712 sensor
- Resistors
- Capacitor
- Potentiometer
- Relay
- Red LED
- Push button
- Lamp (As Load)
- LM044L 20×4 LCD
- AC ammeter
- AC voltmeter
- Virtual Terminal

DESIGN APPROACH

The alternator provides the simulated AC supply. The transformer and voltage-divider circuit provide the voltage-sensing arrangement, while the ACS712 measures load current. The Arduino compares the measured values with the configured protection thresholds and controls the relay to disconnect the lamp when a fault is detected. The LCD displays the measurements and protection status, while the red LED indicates a fault. The Virtual Terminal allows users to modify the protection thresholds, and the reset button clears latched faults.

IMPLEMENTATION

The circuit was assembled and simulated in Proteus 8, and the Arduino firmware was developed to monitor voltage and current. The default overvoltage (OV), undervoltage (UV), and overcurrent (OC) thresholds are 270 V, 207 V, and 2.5 A, respectively. These thresholds can be adjusted through serial commands such as OV=280, UV=200, and OC=2.5, within predefined ranges of 270–300 V, 180–230 V, and 1.5–4.0 A, respectively. The reset button is connected to digital pin 5. The circuit was tested by varying the simulated supply voltage and current to observe the relay, LED, LCD, and reset behavior.

RESULTS

The simulation demonstrated load disconnection under overvoltage, undervoltage, and overcurrent conditions. The LCD displayed the measured values and protection status, and the Virtual Terminal allowed threshold adjustment. The reset button was tested to restore operation after a fault. The project demonstrates the basic protection functions in simulation; physical hardware performance has not been validated.

CONCLUSION

The successful project completion demonstrates the basic functionality of an Arduino-based protection system for overvoltage, undervoltage, and overcurrent conditions in Proteus 8 simulation. All project files, source code, schematics, design documentation, and test evidence are available in the GitHub repository.